

Rethinking real-time collaboration: a local-first approach

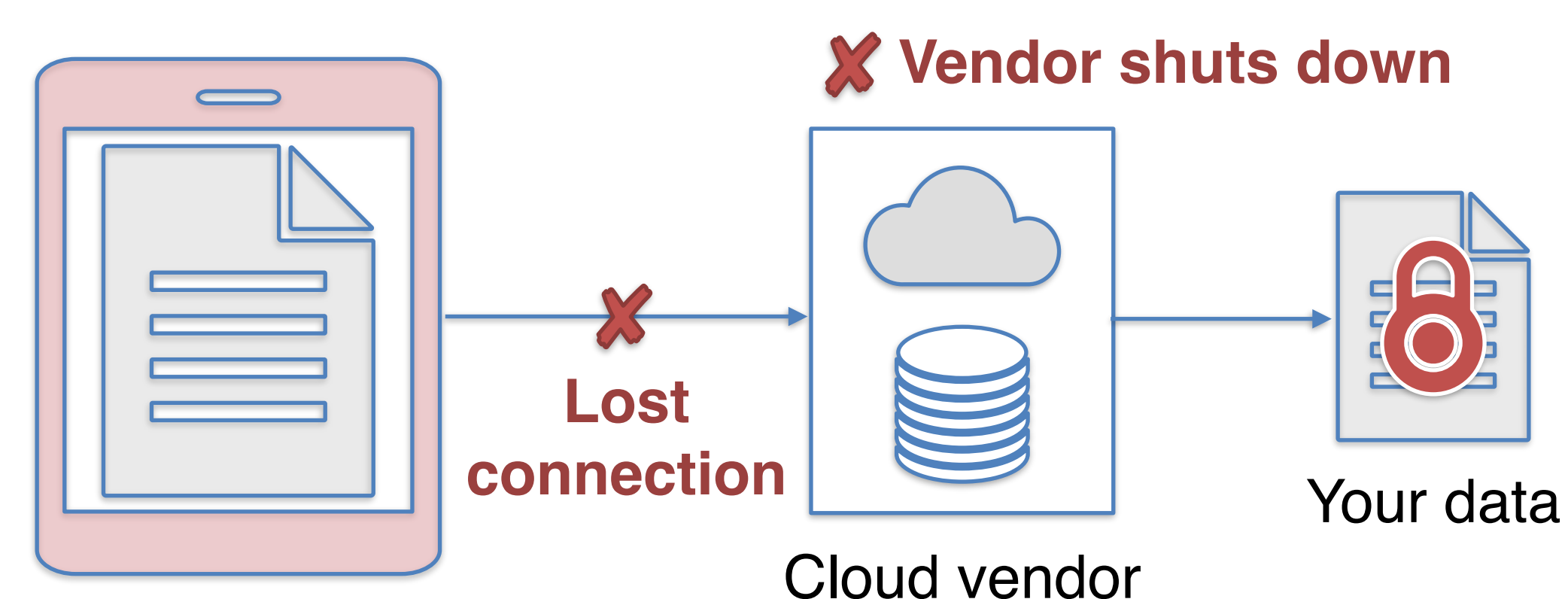
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The problem

Cloud-first software puts your data behind infrastructure you don't control.



Every edit, every sync, and every access request passes through a central server you don't own, and one that can fail, disappear, or lock you out. Even when servers stay online, every keystroke must travel to a data center and back before it is committed, adding latency to every interaction.

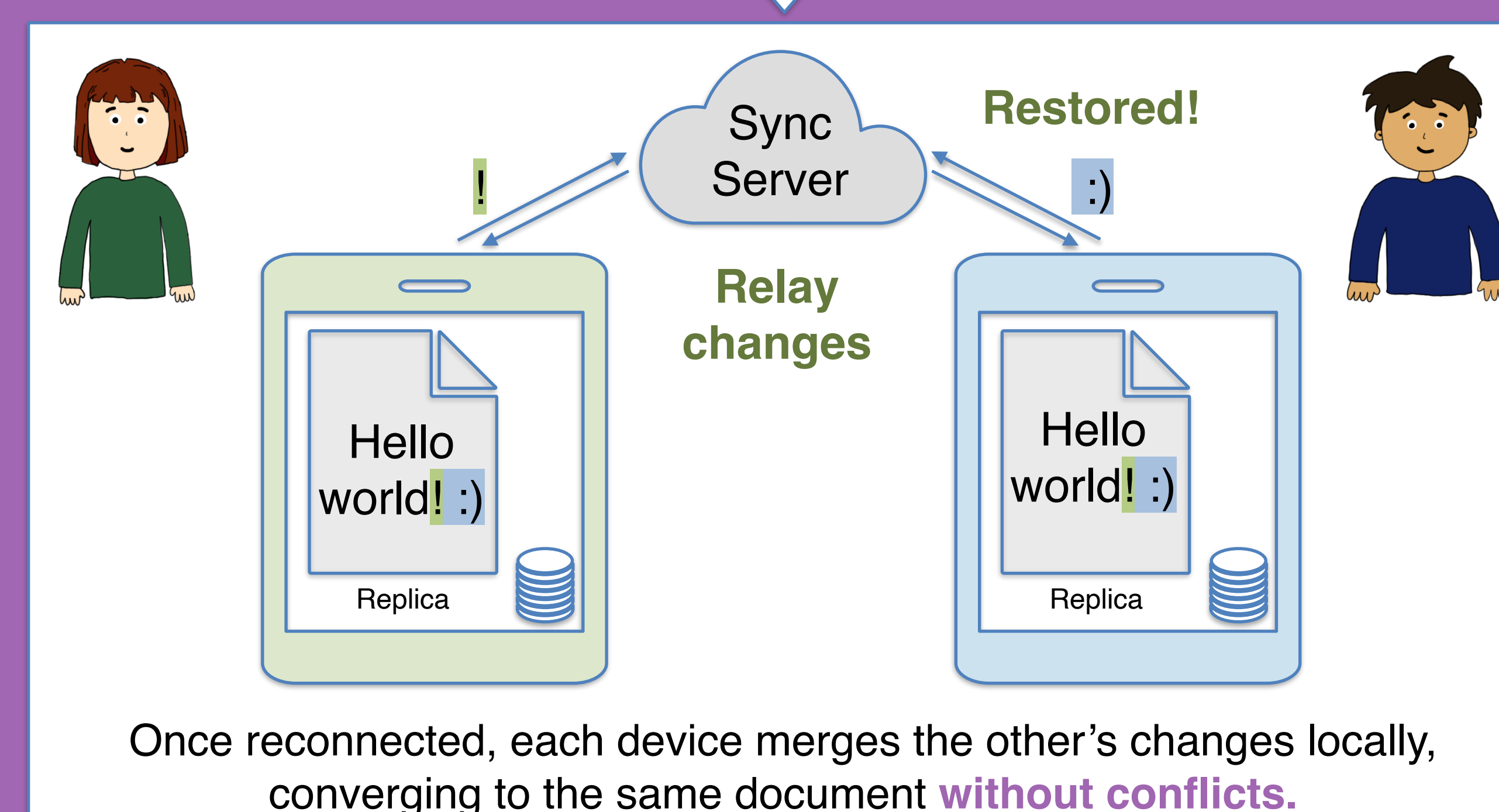
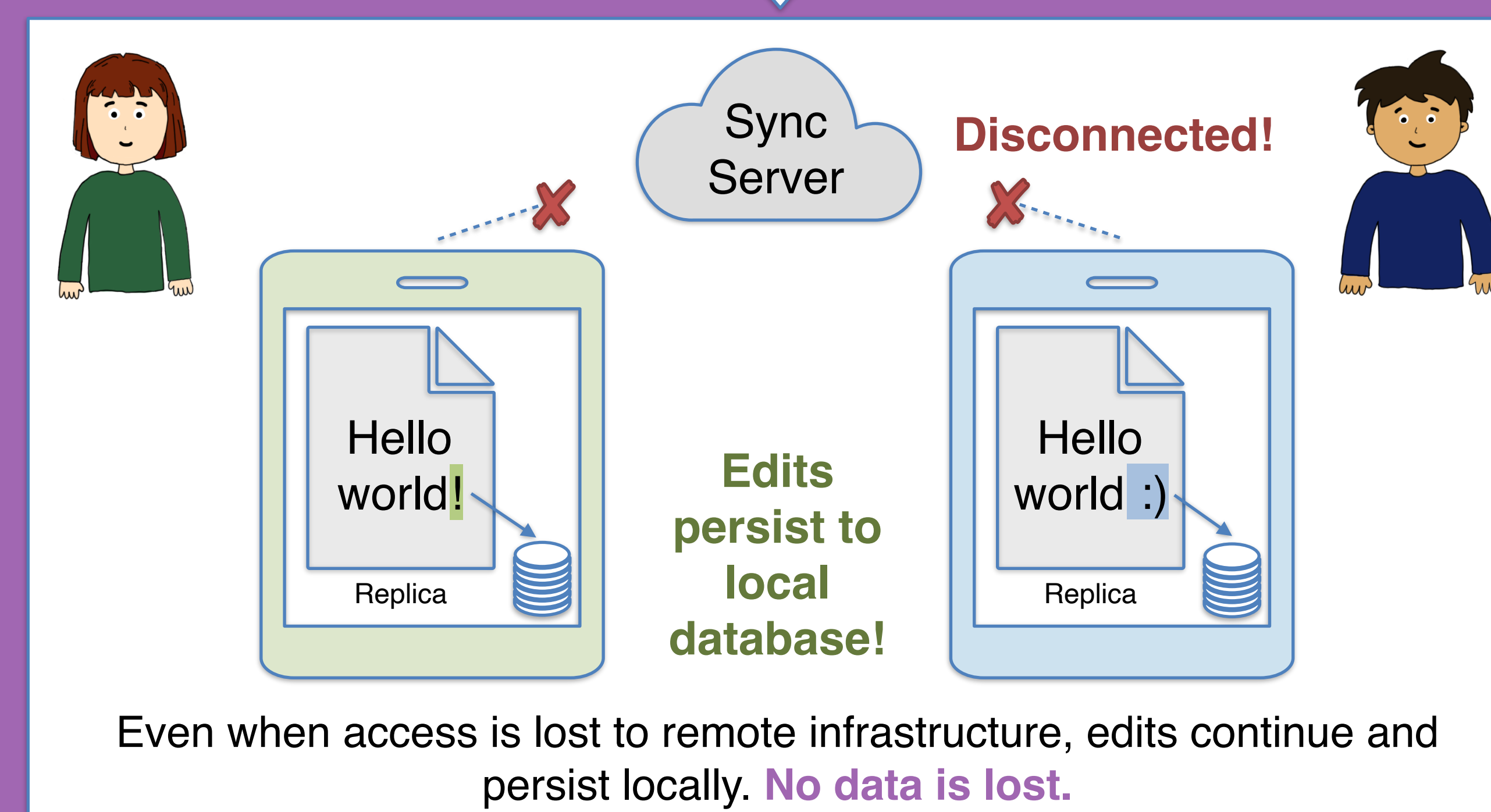
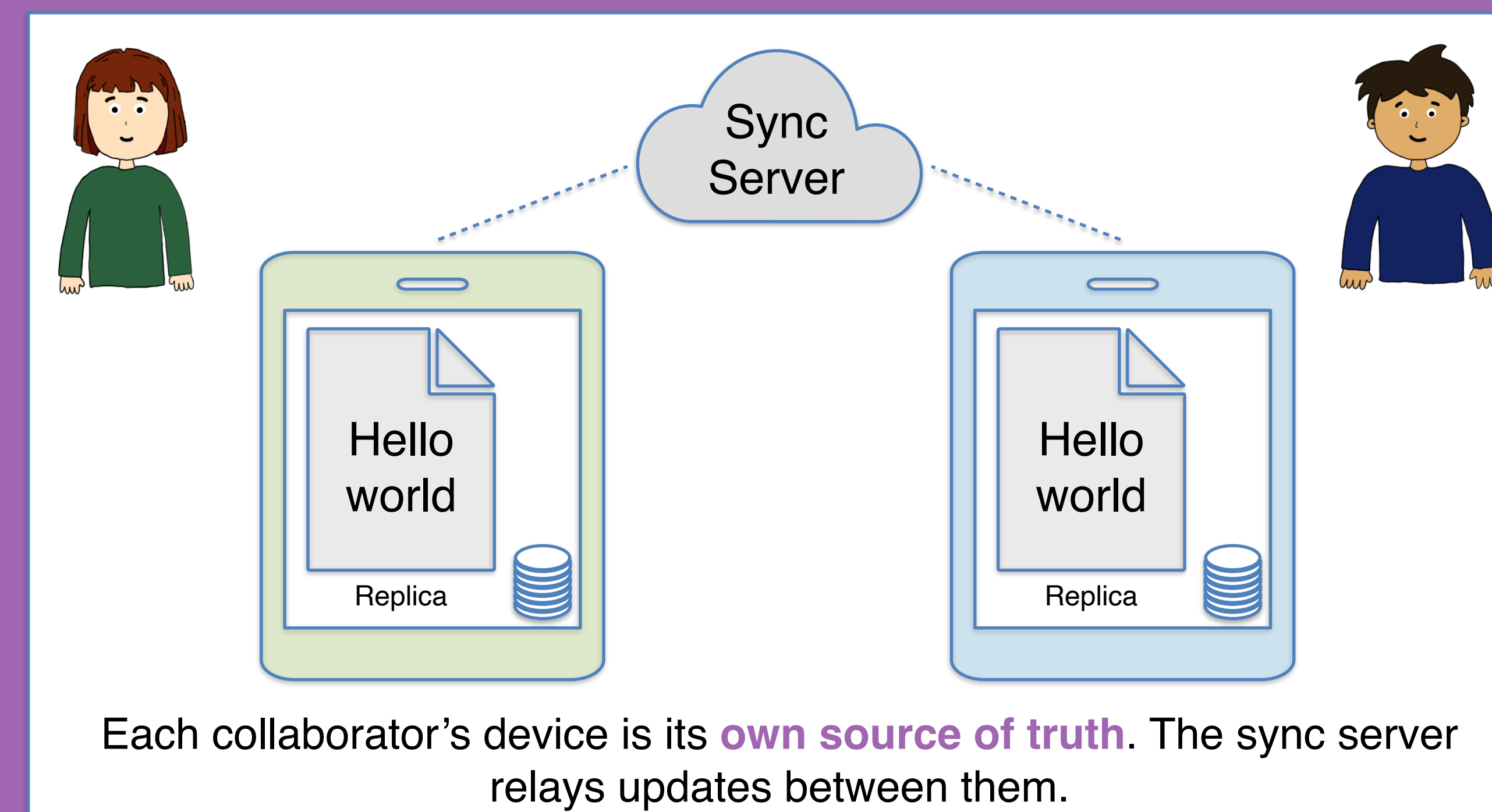
Local-first software in practice

The term “**local-first software**” was first coined by Kleppmann et al. in a 2019 Ink & Switch essay, which laid out a vision for software that gives users ownership and control of their data while still enabling real-time collaboration across multiple devices.

Tools built on these principles include:

- **Yjs**: a high-performance CRDT library, powering collaborative text editors including CodeMirror and ProseMirror
- **Automerge**: a JSON-based CRDT library for building local-first applications across any data type
- **Obsidian**: a note-taking app that stores all data on the user's device

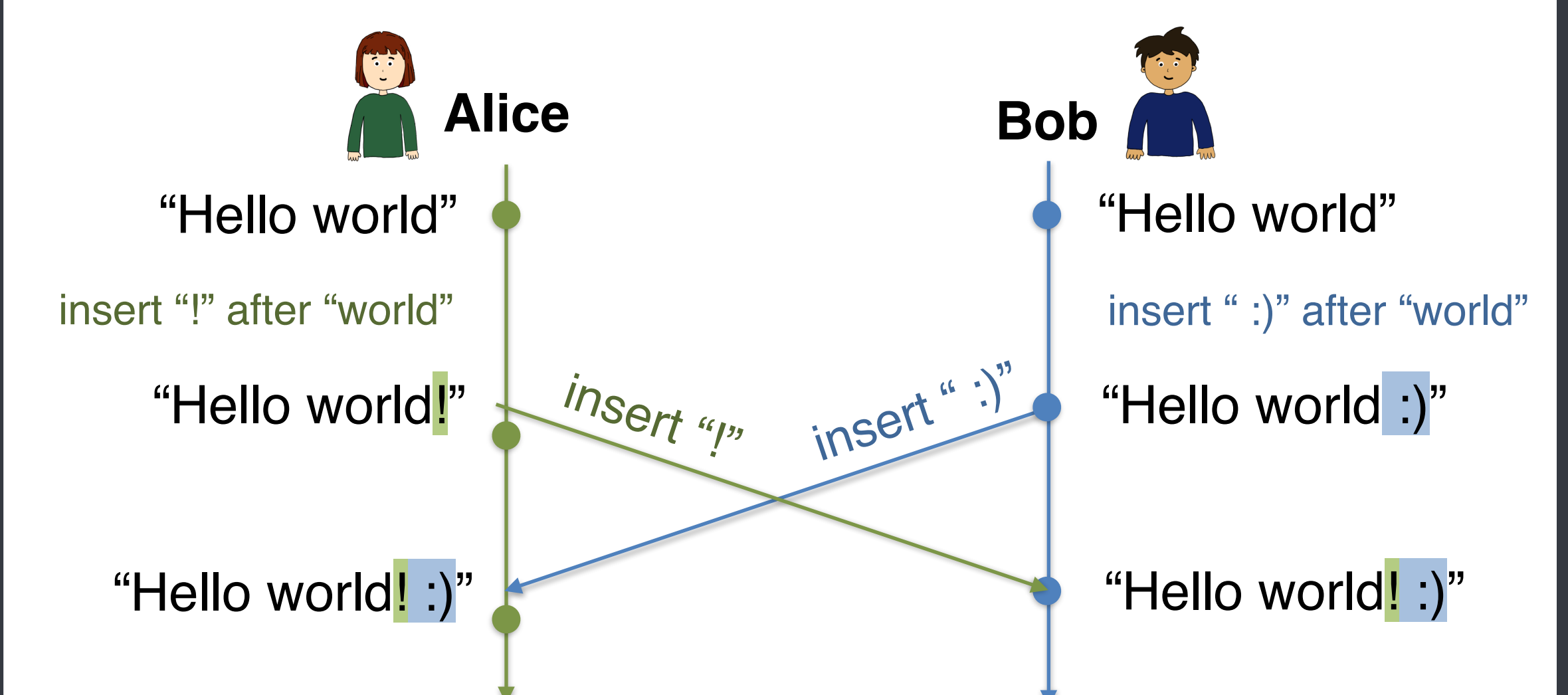
Local-first software enables collaboration that works anywhere and belongs to you.



How does local-first collaboration work?

In local-first software, the primary copy of data lives on each device rather than a remote server. Reads and writes happen locally, sync happens in the background, and **no central server** decides the final state.

The challenge is ensuring every replica converges to the same state without a central arbiter. This is solved by **Conflict-free Replicated Data Types (CRDTs)**, data structures that encode convergence guarantees into their structure. Any two replicas that have seen the same updates will always converge to the same state.



Limitations

- **Conflict complexity**: not all concurrent edits can be resolved automatically
- **Eventual consistency**: collaborators may briefly see different states
- **Data size**: impractical for very large shared datasets

Looking ahead

The cloud made collaboration effortless but quietly transferred ownership of data from users to the companies hosting it. Local-first software demonstrates that seamless collaboration without surrendering that control is possible.

As local-first experiences become more widespread, user expectations will shift accordingly. The path forward requires advances in CRDT algorithms, developer tooling, and the programming models that make local-first software accessible to build.

For references and to learn more about the future of local-first software, scan here:

